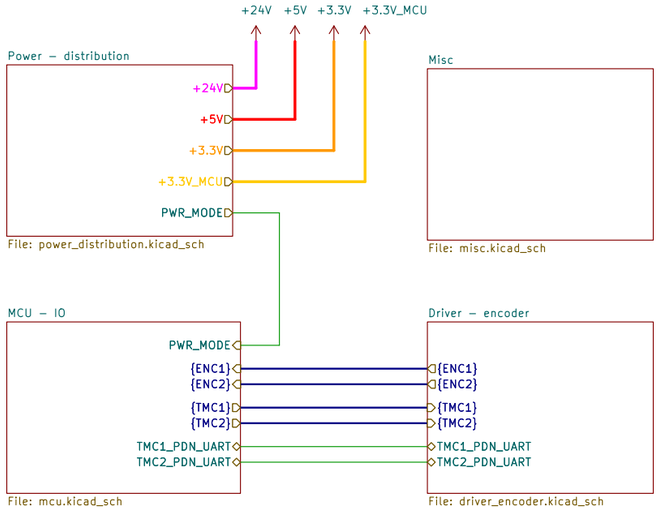
Dual Stepper Driver Evaluation Board

2026-10-22
Rev 1.0



To-Do:

During PCB design phase:
- HSE oscilator
- simulaties
- check USB speed
-> full speed no impedance control

Ordering:
- Check JLCPCB preview PCB

Final design:
- should split IO in seperate sheet
- IPC design rules
- block diagram
- firmware info? State diagram?

#	Test	Procedure	Pass criterion	Result
1	24V supply	24 V only, no motors	+5 V within 4.85–5.15 V, +3.3 V within 3.20–3.40 V, ripple below 50 mVpp	
2	USB supply	USB only, with a C–to–C cable and an A–to–C cable	Board is recognised over USB with both; drivers unpowered	
3	Supply switching	Unplug/replug 24 V while USB is connected, then unplug USB while 24 V is connected; scope +3.3V_MCU	No MCU reset, no USB disconnect, +3.3V_MCU stays above 3.0 V	
4	Reverse polarity	–24 V from a lab supply limited to 0.5 A, for 10 s	Draws under 1 mA; board works afterwards	
5	Hot–plug	Plug in the adapter with the 4 m cable 10 times; scope TP48	Peak below 29 V	
6	Brown–out	Ramp a lab supply 0 → 24 → 0 V slowly, motors connected	No motor movement; clean start and stop	
7	Motor enable	Power up, hold reset, flash firmware, all with 24 V present	No holding torque until the firmware enables the motors	
8	Driver UART	Write and read back GCONF and IHOLD_IRUN on both drivers	Both answer, values match, no error flags	
9	Motor current	Maximum set current; measure across the sense resistor (I = V / 0.22 Ω)	Within ±10 % of the setting and at most 1.0 A RMS	
10	Temperature	Both motors at maximum load for 30 min, board in its enclosure	No overtemperature warning, TMC2209 below 90 °C, other parts below 70 °C	
11	Braking energy	Hardest deceleration at maximum load, and cut 24 V while running	+24 V stays below 29 V	
12	Encoder count	100 revolutions forward and back at maximum speed	Back to exactly 0 counts; one index pulse per revolution; direction correct	
13	Lockstep	Both motors driving the ring gear	Difference between the two encoders below 20 counts	
14	Encoder fault	Unplug an encoder while running	No false counts; firmware flags the fault	
15	USB under load	Stream data for 30 min during maximum acceleration	0 errors, 0 disconnects	
16	Programming	Flash over SWD with the ST–LINK; enter DFU with BOOT0	Both work	
17	ESD	±4 kV contact discharge on the USB shell and the terminals	Keeps working or recovers by itself	

Standard passive component values

If not mentioned otherwise
R: tolerance=1%, package=0603
C: dielectric (not polarized)=X7R, package=0603

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug design notes.

ELECTRICAL CHARACTERISTICS:
Voltage supplied, Current drawn, ...

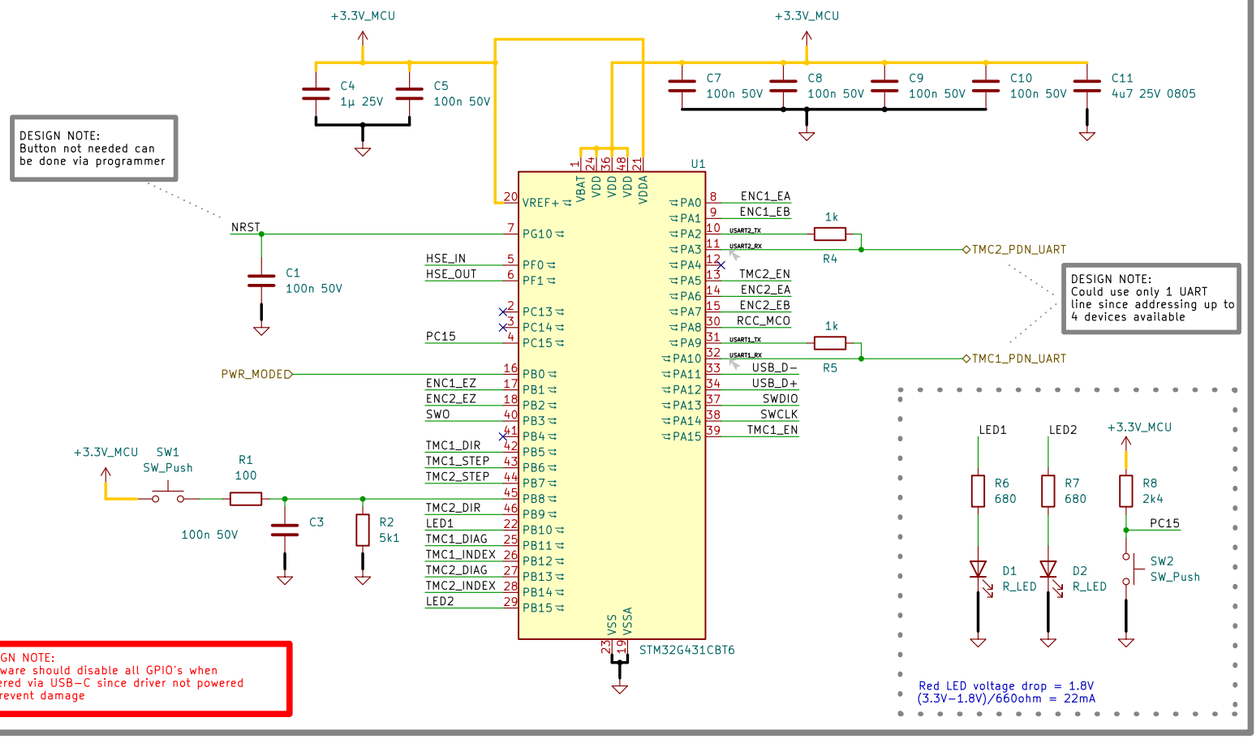
DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

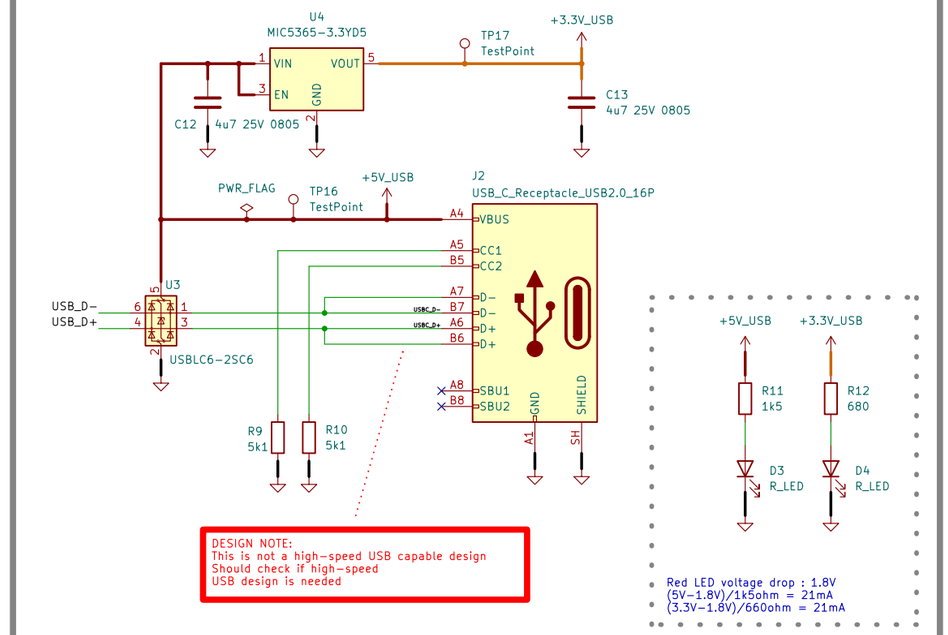
Sheet: /
File: stepper_motor_eval_board_v1.0.kicad_sch
Title: Stepper Motor Evaluation Board
Size: A3 | Date: 2026-10-23 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 1/5

ELECTRICAL CHARACTERISTICS:
3V3 30 mA
-> should be checked with stm tool

MCU



USB



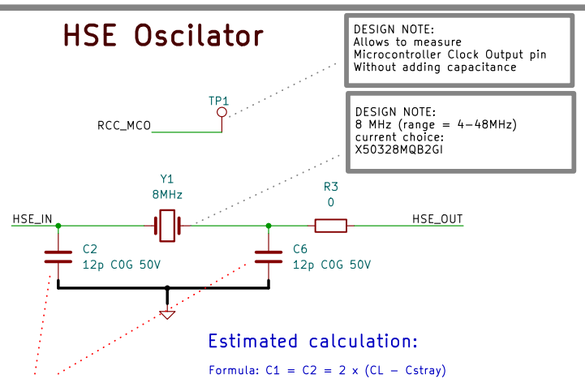
DESIGN NOTE:
No need of LSE oscilator since no sleep activity

DESIGN NOTE:
HSE oscilator since space and BOM allows it
Better accuracy timing for everything including UART and USB

DESIGN NOTE:
There are no constraints on crystal size, then it is recommended to choose a crystal with the smallest possible ESR.

DESIGN NOTE:
If there are no constraints on power consumption, it is recommended to choose crystals with high load capacitance.

HSE Oscilator



DESIGN NOTE:
Allows to measure Microcontroller Clock Output pin Without adding capacitance

DESIGN NOTE:
8 MHz (range = 4-48MHz) current choice: X50328MQB2GI

DESIGN NOTE:
Capacitors values should still be calculated depending on PCB traces etc.

Use 0402 footprint for easy replacement of caps when testing

Estimated calculation:

Formula: C1 = C2 = 2 x (CL - Cstray)

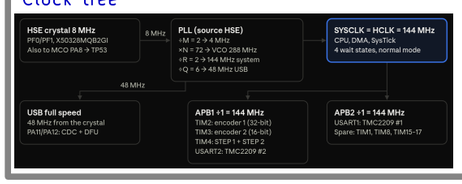
with Cstray estimated = 10pF p.100 datasheet MCU
C1 and C2 should be between 5pF-20pf

X50328MQB2GI:
CL = 16pF -> so that C1, C2 are in middle of allowed range
ESR = 80ohm
Tolerance = +-10ppm
Drive level = 100µW or higher

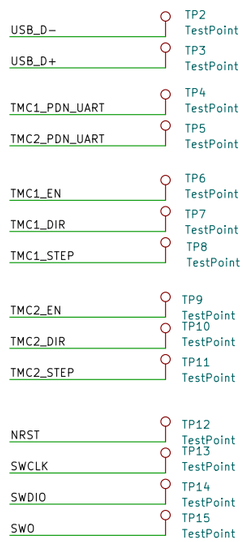
C1 = C2 = 12pF (estimate) -> good in middle of range

LAYOUT NOTE:
- place crystal and caps very close to OSC pins
- keep traces short and symmetrical
- avoid noisy digital routing near the oscillator loop
- avoid excessive vias in crystal traces
- respect the datasheet's ESR and drive recommendations

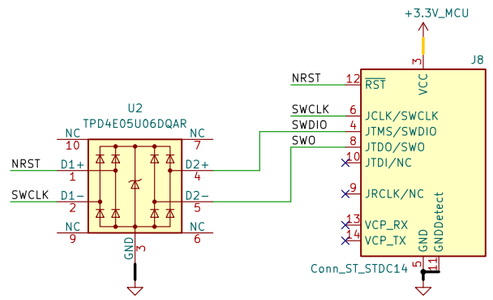
Clock tree



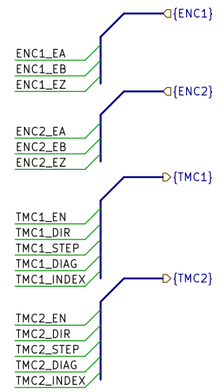
Testing points



SWD (Debug) Port



Interconnections



Sheet: /MCU - IO/
File: mcu.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/5

Stepper motor drivers

ELECTRICAL CHARACTERISTICS:
Driver current depending on motor: 1A/phase with 2ohm phase resistance
Pmotor_coil = (1A)² x (2ohm) = 2W
Motor has two coils so : 2W x 2 = 4W per motor
Idriver = P/U = 4W / 24V = 0.167A
Two drivers in total so : 0.167A x 2 = 0.33A

Two drivers = 24V 0.5A

One driver IO = 3V3 30µA
Two driver IO = 3V3 60µA

DESIGN NOTE:
Should still check layout example p82

DESIGN NOTE:
Do not power MCU output pin when IC not powered= when MCU only powered by USB

DESIGN NOTE:
Aluminium electrolytic capacitor

LAYOUT NOTE:
Place near IC with short path to die pad

DESIGN NOTE:
Chopper mode selection done via UART

DESIGN NOTE:
Diagnostic and StallGuard output.
Hi level upon stall detection or driver error.
Diagnostic done via UART

DESIGN NOTE:
In combination with a mechanical home switch, a more precise homing is enabled.
see p.8 and p66

DESIGN NOTE:
Oscilloscope testpoints for voltage measure of stepper phases

DESIGN NOTE:
Oscilloscope testpoints for current sense through phases of stepper

LAYOUT NOTE:
Connect directly to GND plane and use low inductivity SMD type

DESIGN NOTE:
p.52 current sense resistor of 220mOhm for 1A motor

DESIGN NOTE:
Do not hot plug motor connector

{TMC1} TMC1_STEP TMC1_DIR TMC1_EN TMC1_DIAG TMC1_INDEX TMC1_PDN_UART

{TMC2} TMC2_STEP TMC2_DIR TMC2_EN TMC2_DIAG TMC2_INDEX TMC2_PDN_UART

+3.3V +24V

C15 100n 50V · C16 100n 50V · C17 100n 50V · C20 100n 50V · C21 100µ 50V · D6 SMAJ24A

C23 100n 50V · C24 100n 50V · C25 100n 50V · C28 100n 50V · C29 100µ 50V · D8 SMAJ24A

U7 / U10 TMC2209-LA

STEP DIR CLK PD/UART MS1/AD0 MS2/AD1 SPREAD EN STDBY DIAG INDEX VCC_IO VS VCP 5VOUT VREF CPO CPI OA1 OA2 OB1 OB2 BRA BRB GND NC

C19 2.2µ 25V · C27 2.2µ 25V · C18 22n 50V · C26 22n 50V

R13 10k · R17 10k

R14 680 · R18 680

R15 220m 2010 · R16 220m 2010 · R19 220m 2010 · R20 220m 2010

Red LED voltage drop = 1.8V (3.3V−1.8V)/660ohm = 22mA

D5 R_LED · D7 R_LED

J4 Motor1_Conn · J6 Motor2_Conn

TP27 TP28 TP29 TP30 TP31 TP32 TP42 TP43 TP44 TP45 TP46 TP47

Encoder – RS422 differential signals to logic signals

ELECTRICAL CHARACTERISTICS
1 AM26LV32E :
3V3 17mA max supply current
2 AM26LV32E :
3V3 17mA x 2 = 34mA

ELECTRICAL CHARACTERISTICS:
1 Encoder :
5V 50mA estimated, to be measured
2 Encoders :
5V 50mA x 2 = 100mA

LAYOUT NOTE:
Should check to add termination resistance based on simulation

{ENC1} ENC1_EA ENC1_EB ENC1_EZ

{ENC2} ENC2_EA ENC2_EB ENC2_EZ

+3.3V +5V

C14 100n 50V · C22 100n 50V

U5 / U8 AM26LV32EIDRG4

1A 1B 2A 2B 3A 3B 4A 4B 1Y 2Y 3Y 4Y G G̅ VDD GND

U6 / U9 TPD6E05U06RVZR

D1+ D1− D2+ D2− D3+ D3− GND NC

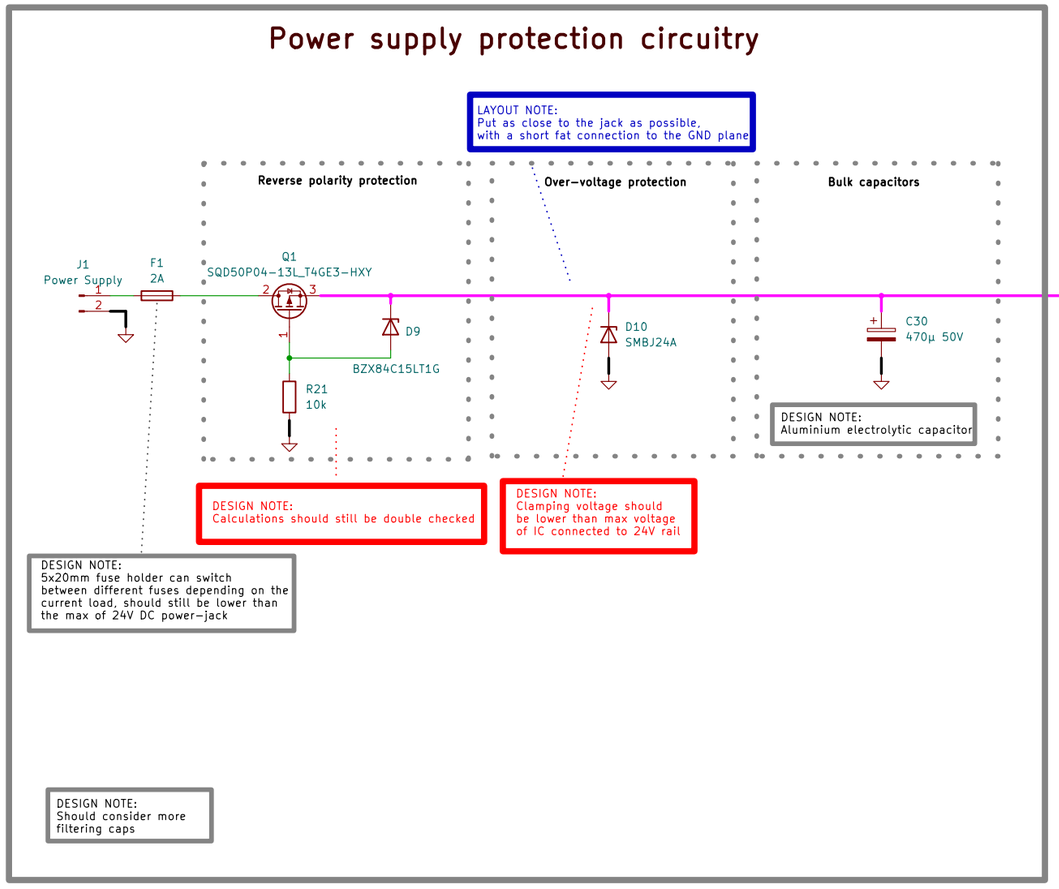
J3 Encoder1_Conn · J5 Encoder2_Conn

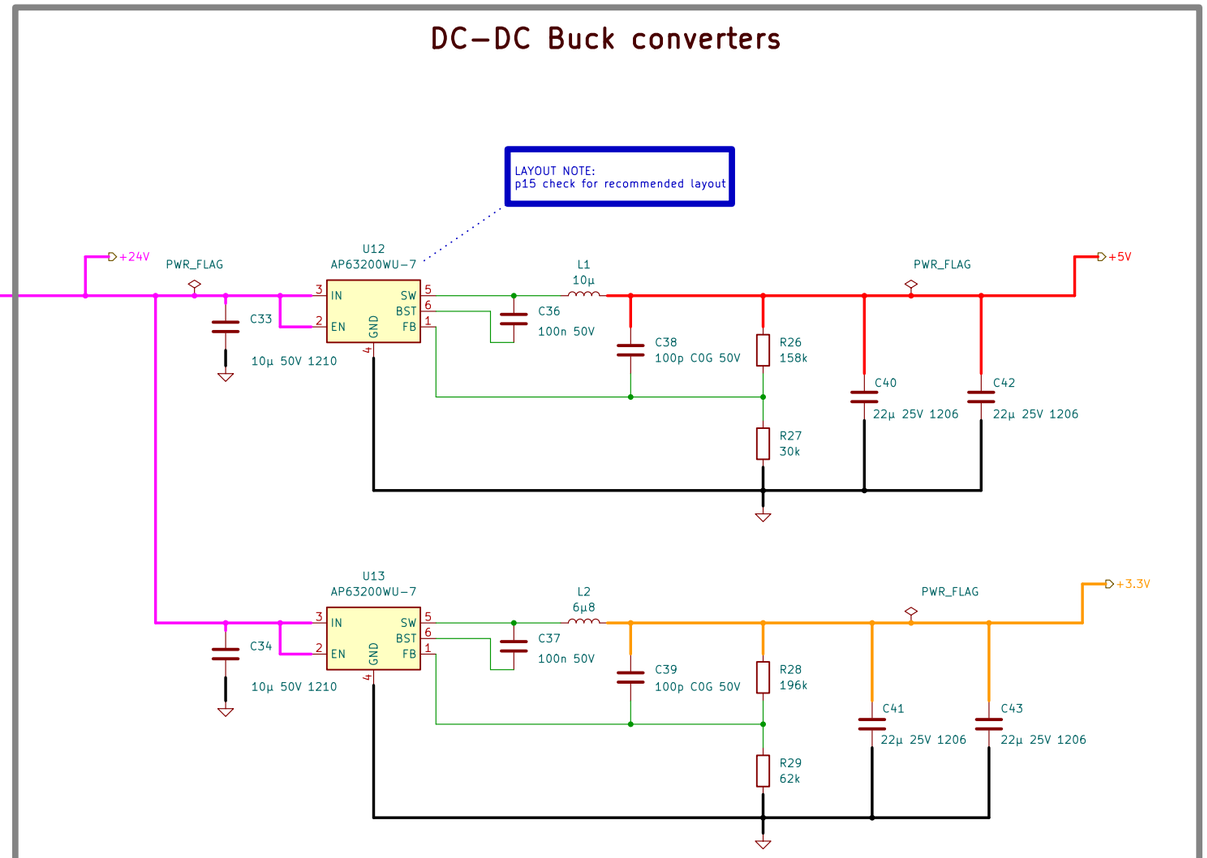
TP18 TP19 TP20 TP21 TP22 TP23 TP24 TP25 TP26 TP33 TP34 TP35 TP36 TP37 TP38 TP39 TP40 TP41

Sheet: /Driver – encoder/
File: driver_encoder.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 3/5





Current consumption

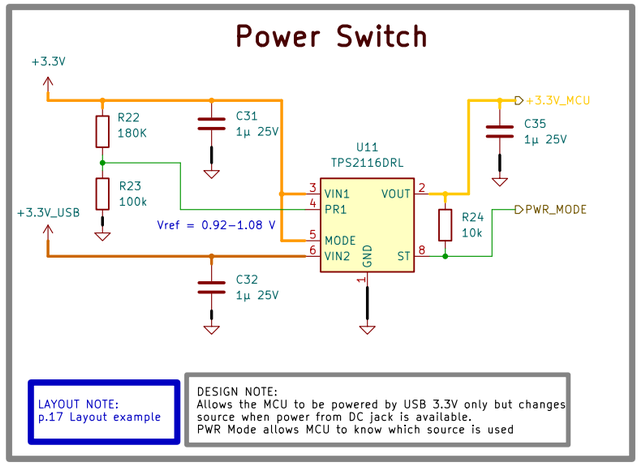
Voltage rail	Components	Estimated current draw	Dimensioned current draw
24V	2x motor (driver)	0.5A	2A
5V	2x encoders	0.1A	1A
3V3	2x driver IC, RS422 signal converter	60µA + 34mA = 0.03406A	0.5A
5V_USB	MCU	30mA	0.5A

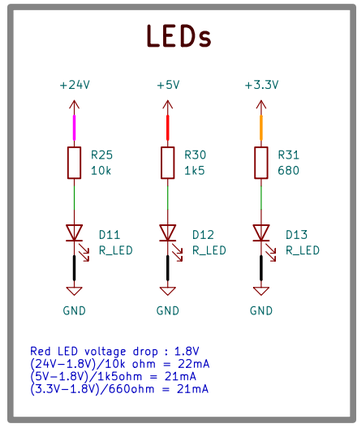
Component selector DC-DC buck converter

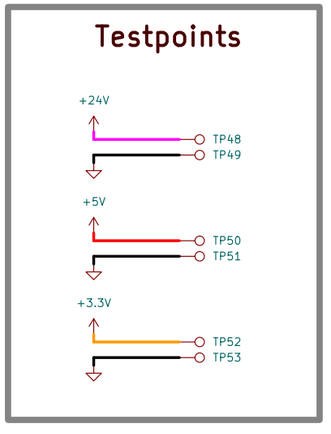
Table 1 shows a list of recommended component selections for common output voltages for the AP63200 and AP63201 referencing Figure 20.

AP63200/AP63201							
Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	L (µH)	C1 (µF)	C2 (µF)	C3 (nF)	C4 (pF)
1.2	30.9	62	2.2	10	2 x 22	100	100
1.5	54.2	62	2.2	10	2 x 22	100	100
1.8	77.5	62	3.3	10	2 x 22	100	100
2.5	131	62	3.3	10	2 x 22	100	100
3.3	196	62	6.8	10	2 x 22	100	100
5	157	30	10	10	2 x 22	100	100
12	249	18	10	10	2 x 22	100	56

Table 1. Recommended Component Selections for AP63200/AP63201





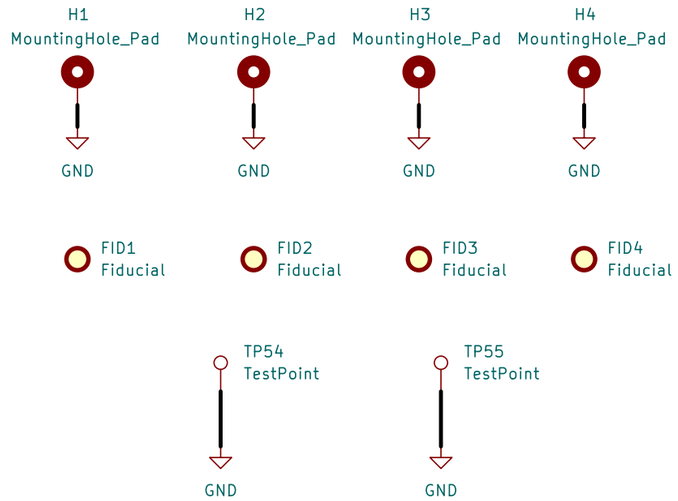


Sheet: /Power - distribution/
File: power_distribution.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 4/5

Misc



Sheet: /Misc/
File: misc.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/5